Solution

Q.1 $(395.4)_{10} = (12023.12)_4$

a)

4	395	
4	98	3
4	24	2
4	6	0
	1	2

$$\begin{array}{r} .4 \\ \times 4 \\ \hline 1.6 \\ \times 4 \\ \hline 2.4 \end{array}$$

$(1\ 2\ 0\ 2\ 3 . 1\ 2)_4$

$(\underbrace{01\ 10}\ \underbrace{00\ 10\ 11}\ .\ \underbrace{01\ 10})_2$

$(6\ \ 1\ \ 3\ .\ 3)_8$

Q.1

b) $(1\ 1\ 2\ 2 . 1)_3$

$1\times3^3 + 1\times3^2 + 2\times3^1 + 2\times3^0 + 1\times3^{-1}$

$= 1\times27 + 1\times9 + 2\times3 + 2\times1 + 1\times\frac{1}{3}$

$= 27 + 9 + 6 + 2 + \frac{1}{3}$

$= 34.33$

or $34\frac{1}{3}$

$$\begin{array}{r} .33 \\ 3\overline{)10} \\ -9 \\ \hline 10 \\ -9 \end{array}$$

$$\begin{array}{r} 27 \\ 9 \\ 6 \\ +\ 2 \\ \hline 34 \end{array}$$

Solution

Q.2 a)

$$(A+B)(A+C)(\bar{A}+\bar{B}+\bar{C}) = A \oplus BC$$

$$L.H.S = (A+B)(A+C)(\bar{A}+\bar{B}+\bar{C})$$

$$= (A+B)(0+A\bar{B}+A\bar{C}+\bar{A}C+\bar{B}C+0)$$

$$= (A+B)(A\bar{B}+A\bar{C}+\bar{A}C+\bar{B}C)$$

$$= A\bar{B}+A\bar{C}+0+A\bar{B}C+0+AB\bar{C}+\bar{A}BC+0$$

$$= A(\bar{B}+\bar{C})+A(\bar{B}C+B\bar{C})+\bar{A}BC$$

$$= A(\bar{B}+\bar{C}+\bar{B}C+B\bar{C})+\bar{A}BC$$

$$= A(\bar{B}(1+C)+\bar{C}(1+B))+\bar{A}BC$$

$$= A(\bar{B}+\bar{C})+\bar{A}BC$$

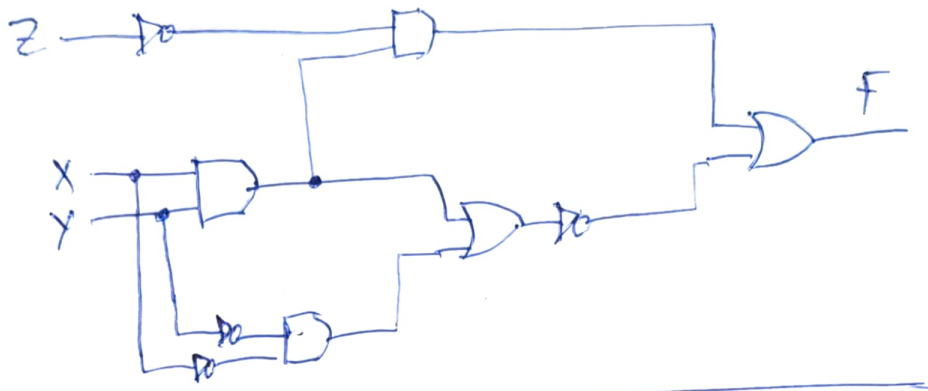
$$= A(\overline{BC})+\bar{A}(BC)$$

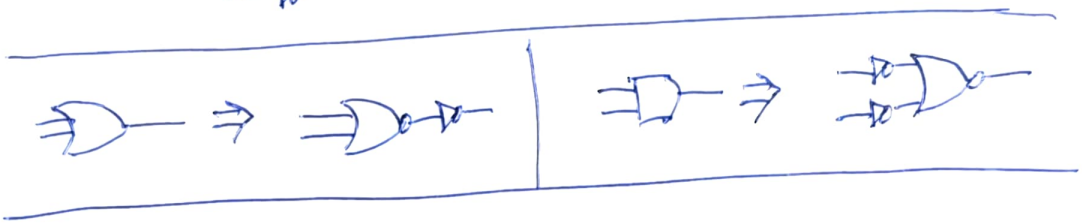
$$= A \oplus BC$$

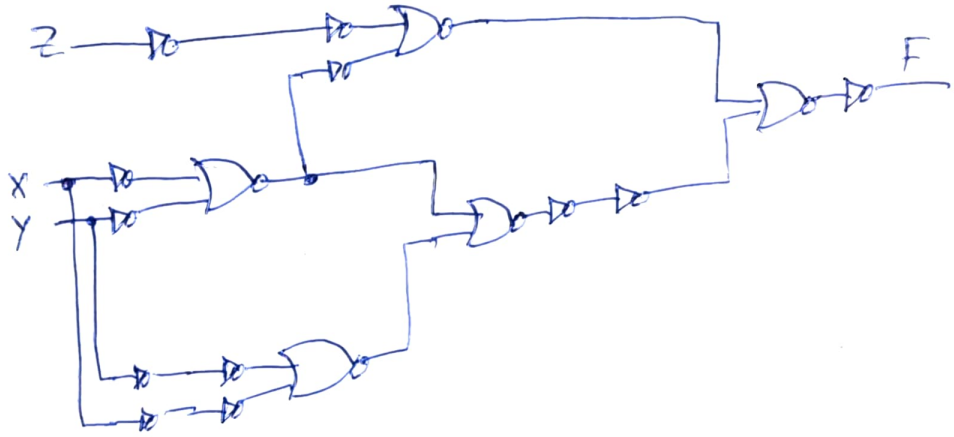
$$= R.H.S$$

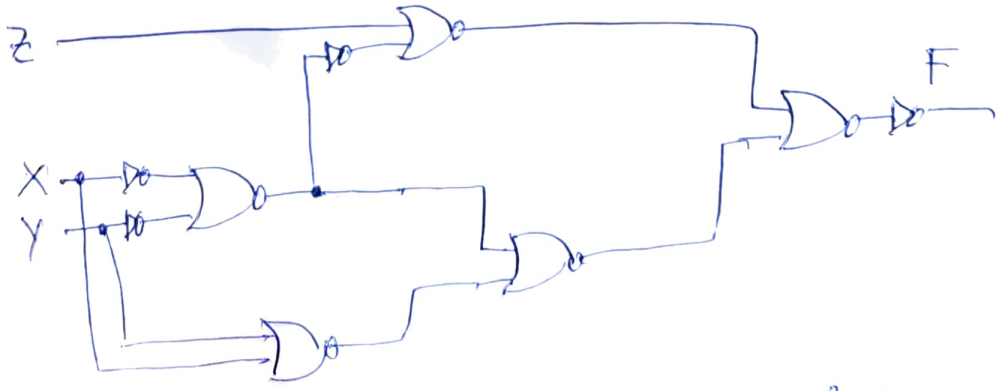
Solution

Q.2
b)









Final answer

Q.3 a)

Solution

$F(A,B,C,D) = \Pi_M(1,3,6,7,8,9,14,15)$

$d(A,B,C,D) = \Pi_M(0,11,13)$

F

AB \ CD	00	01	11	10
00	X			1
01	1	1		
11	1	X		
10			X	1

SOP

$F = B\bar{C} + \bar{B}C\bar{D}$

F

AB \ CD	00	01	11	10
00	X	0	0	
01			0	0
11		X	0	0
10	0	0	X	

POS

$\bar{F} = \bar{B}\bar{C} + CD + BC$

$F = (B+C)(\bar{C}+\bar{D})(\bar{B}+\bar{C})$

Q.3 b)

$F = \Sigma m(1,2,4,7,8,11,13,14)$

A B C D	F
0 0 0 0	0
0 0 0 1	1
0 0 1 0	1
0 0 1 1	0
0 1 0 0	1
0 1 0 1	0
0 1 1 0	0
0 1 1 1	1
1 0 0 0	1
1 0 0 1	0
1 0 1 0	0
1 0 1 1	1
1 1 0 0	0
1 1 0 1	1
1 1 1 0	1
1 1 1 1	0

$F = \bar{A}\bar{B}\bar{C}D + \bar{A}\bar{B}C\bar{D} + \bar{A}B\bar{C}\bar{D} + \bar{A}BCD$

$+ A\bar{B}C\bar{D} + A\bar{B}CD + AB\bar{C}D + ABC\bar{D}$